Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

The half-life of a substance is 20 minutes. What is the time interval between 33% decay and 67% decay?

(A) $40\ \text{min}$

(B) $20\ \text{min}$

(C) $30\ \text{min}$

(D) $25\ \text{min}$

Q.2

The half life of an element A is equal to the mean life of an element B. If, initially (i.e., at $t = 0$), A and B have equal number of radioactive nuclei, then

(A) initially both A and B decay at the same rate

(B) A and B always decay at the same rate

(C) A decays at a faster rate than B

(D) B decays at a faster rate than A.

Q.3

Two radioactive materials A and B have decay constant $\frac{13}{7}$ units and $\frac{19}{14}$ units respectively. Directly both have the same number of nuclei. The time (in same system of units), after which the ratio of their remaining nuclei will be $1/\mathrm{e}$ is

(A) 2

(B) 3

(C) 4

(D) 5

Q.4

A radioactive sample contains $10^{-3}\ \text{kg}$ each of two nuclear species A and B with half-life 4 days and 8 days respectively. The ratio of the amounts of A and B after a period of 16 days is

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(A) $1:2$

(B) $4:1$

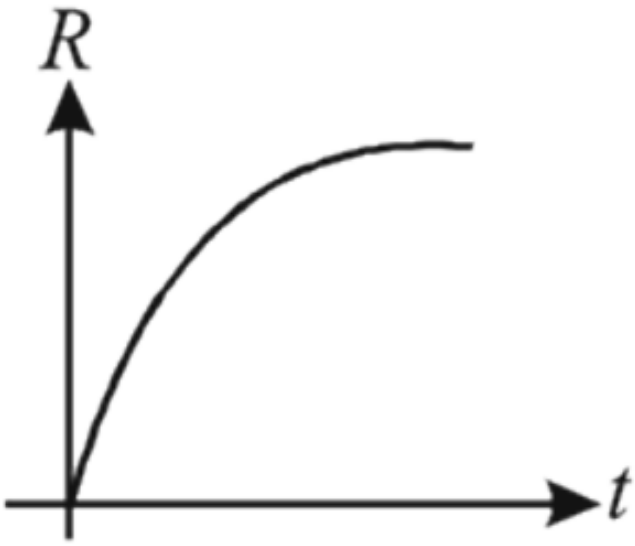
(C) $1:4$

(D) $2:1$

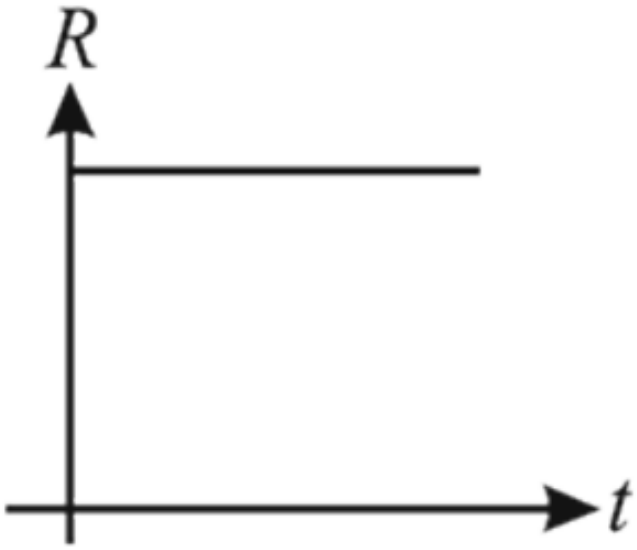
Q.5

A radioactive nucleus X decays to a stable nucleus ' Y. Then the graph of rate of formation of $'Y'$ against time t ' will be

(A)

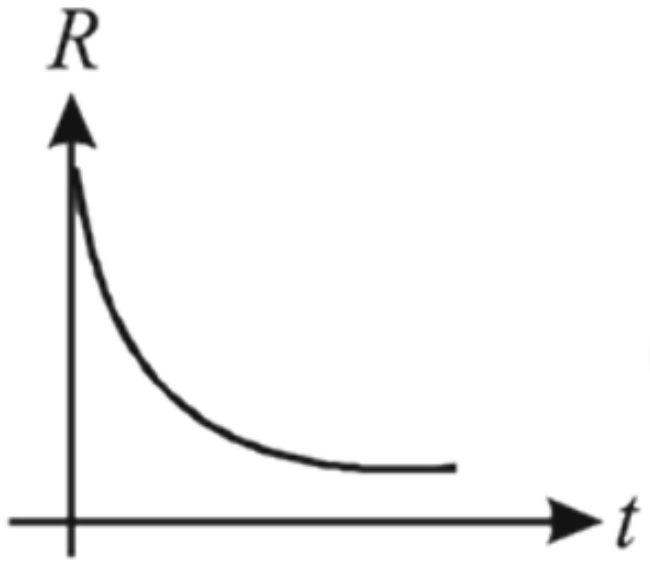


(B)

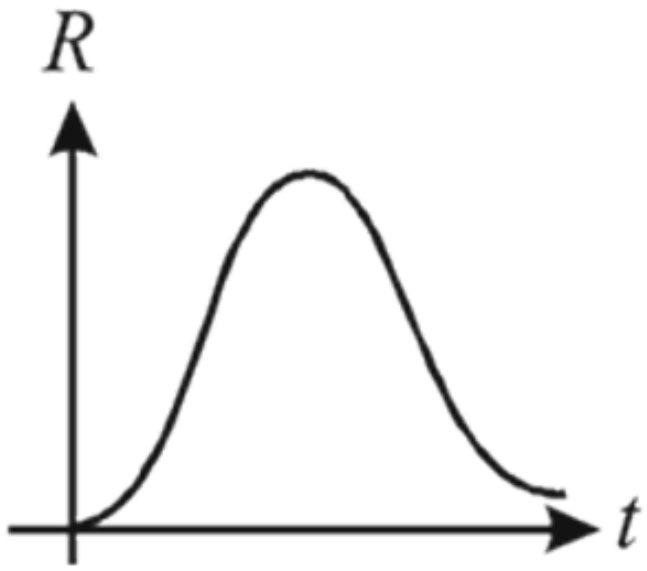


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(C)



(D)



Q.6

A radioactive material of half life $\ln 2$ was produced in a nuclear reactor. Consider two different instants A and B. The number of undecayed nuclei at instant B was twice of that of instant A. If the activities at instants A and B are A_1 and A_2 respectively then the difference in the age of the sample at these instants equals.

(A) $\left|\ln\left(\frac{2\,\mathrm{A}_1}{\mathrm{A}_2}\right)\right|$

(B) $\ln 2\left|\ln\left(\frac{\mathrm{A}_1}{\mathrm{A}_2}\right)\right|$

(C) $\left|\ell\mathrm{n}\left(\frac{\mathrm{A}_2}{2\,\mathrm{A}_1}\right)\right|$

(D) $\ln 2\left|\ln\left(\frac{\mathrm{A}_2}{\mathrm{A}_1}\right)\right|$

Q.7

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An archaeologist analyses the wood in a prehistoric structure and finds that C^{14} (Halflife $= 5700$ years) to C^{12} is only one-fourth of that found in the cells of buried plants. The age of the wood is about

(A) 5700 years

(B) 2850 years

(C) 11,400 years

(D) 22,800 years

Q.8

Half lives for α and β emission of a radioactive material are 16 years and 48 years respectively. When material decays giving α and β emission simultaneously, time in which $3/4^{\text{th}}$ material decays is

(A) 29 years

(B) 24 years

(C) 64 years

(D) 12 years

Q.9

A radioactive source in the form of metal sphere of diameter 10^{-3} m emits beta particle at a constant rate of 6.25×10^{10} particles per second. If the source is electrically insulated, how long will it take for its potential to rise by 1.0 volt, assuming that 80% of the emitted beta particles
escape from the source?

(A) 6.95μsec

(B) 0.95μsec

(C) 1.95μsec

(D) 2.15μsec

Q.10

A radioactive nucleus undergoes a series of
decay according to the scheme

$$A \xrightarrow{a} A_1 \xrightarrow{\beta} A_2 \xrightarrow{\alpha} A_3 \xrightarrow{\gamma} A_4$$

If the mass number and atomic number of 'A' are 180 and 72 respectively, then what are these numbers for A_4

(A) 172 and 69

(B) 174 and 70

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(C) 176 and 69

(D) 176 and 70

Q.11

Consider α particles, β particles and γ - rays, each having an energy of $0.5\mathrm{MeV}$. In increasing order of penetrating powers, the radiations are:

(A) α, β, γ

(B) α, γ, β

(C) β, γ, α

(D) γ, β, α

Q.12

At radioactive equilibrium, the ratio between the atoms of two radioactive elements (X) and (Y) was found to be $3.2 \times 10^9 : 1$ respectively. If half-life of the element (X) is 1.6×10^{10} years, then half-life of
the element (Y) would be

(A) 3.2×10^9 years

(B) 5×10^9 years

(C) 1.6×10^{10} years

(D) 5 years

Q.13

A radioactive element X converts into another stable element Y. Half life of X is $2\mathrm{hrs}$. Initially only X is present. After time t, the ratio of atoms of X and Y is found to be $1 : 4$, then t in hours is

(A) 2

(B) 4

(C) between 4 and 6

(D) 6

Q.14

For a radioactive sample the counting rate changes from 6520 counts/minute to 3260 counts / minute in 2 minutes. Determine the decay constant.

(A) 1.78 per sec

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(B) 0.78 per sec

(C) 2.78 per sec

(D) 5.78 per sec

Q.15

Consider a radioactive material of half-life 1.0 minute. If one of the nuclei decays now, the next one will decay

(A) after 1 minute

(B) after $\frac{1}{\log_e 2}$ minute

(C) after $\frac{1}{N}$ minute, where N is the number of nuclei present at that moment

(D) after any time

Q.16

After 150 days, the activity of a radioactive sample is 5000 dps. The activity becomes 2500 dps after another 75 days. The initial activity of the sample is

(A) 20000dps

(B) 40000dps

(C) 7500dps

(D) 10000dps

Q.17

If a radioactive material contains 0.1mg of Th^{234} how much of it will remain unchanged after 120 days. Given Half life is 24 days

(A) $1.12\mu g$

(B) $0.1\mu g$

(C) $3.125\mu g$

(D) $125\mu g$

Q.18

Which of the following statements is false:

(A) The energy spectrum of $\alpha - \text{decay}$ is line spectrum.

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(B) The energy spectrum of $\beta-$ particles is continuous.

(C) The energy spectrum of $\gamma-$ decay is continuous.

(D) The energy spectra of α & $\gamma-$ decay are line spectra.

Q.19

The rate of disintegration of fixed quantity of a radioactive element can be increased by

(A) Increasing the temperature

(B) Increasing the pressure

(C) Chemical reaction

(D) It is not possible

Q.20

A radioactive nucleus can decay by two different processes. The average life for the first process is t_1 and that for the second process is t_2. The effective average life for the two processes is :

(A) $\frac{t_1+t_2}{2}$

(B) $t_1 + t_2$

(C) $\sqrt{t_1 t_2}$

(D) $\frac{t_1 t_2}{t_1+t_2}$

Answers & Solutions

Q.1 Answer:

20 min

Solution:

Given $T = 20$ min. **We know that**

$\frac{N}{N_0} = \left(\frac{1}{2}\right)^{t/T}$

For 33% **decay,** $\frac{N}{N_0} = \frac{100-33}{100} = \frac{67}{100}$

If t_1 **is the time for** 33% **decay, then**

$\frac{67}{100} = \left(\frac{1}{2}\right)^{t_1/T}$

For 67 % decay, $\frac{N}{N_0} = \frac{100-67}{100} = \frac{33}{100}$

If t_2 **is the time for** 67% **decay, then**

$\frac{33}{100} = \left(\frac{1}{2}\right)^{t_2/T}$

Dividing (ii) by (i), we have

$\frac{33}{67} = \left(\frac{1}{2}\right)^{(t_2-t_1)/T}$

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Now $\frac{33}{67} \simeq \frac{1}{2}$. Hence Eq. (iii) reduces to

$\left(\frac{1}{2}\right)^1 = \left(\frac{1}{2}\right)^{(t_2-t_1)/T}$

which gives $1 = \frac{t_2-t_1}{T}$ **or** $t_2 - t_1 = T = 20$ min.

Hence the correct choice is (b).

Q.2 Answer:

B decays at a faster rate than A.

Solution:

Half life $T = \frac{\ln(2)}{\lambda}$

Mean life $\tau = \frac{1}{\lambda}$**. Therefore**

$T = \tau \ln(2)$

$T_A = \tau_{\mathrm{A}} \ln(2)$

$T_B = \tau_{\mathrm{B}} \ln(2)$

Given $T_A = \tau_{\mathrm{B}}$**. Hence** $\frac{T_B}{T_A} = \ln(2) = 0.693$

So $T_B < T_{\mathrm{A}}$**, which implies** $\lambda_{\mathrm{B}} > \lambda_{\mathrm{A}}$

since $\frac{dN}{dt} = -\lambda \mathrm{N}$

$\frac{dN_A}{dt} = -\lambda_{\mathrm{A}} N_{\mathrm{A}}$

and $\frac{dN_B}{dt} = -\lambda_{\mathrm{B}} N_{\mathrm{B}}$

Given $N_{\mathrm{A}} = N_{\mathrm{B}}$**. Hence** $= \frac{\frac{dN_A}{dt}}{\frac{dN_B}{dt}} = \frac{\lambda_A}{\lambda_B} < 1$

or $\frac{dN_B}{dt} > \frac{dN_A}{dt}$

Hence B **decays at a faster rate then** A**. So the correct choice is (d).**

Q.3 Answer:

2

Solution:

$\frac{\mathrm{N}_1}{\mathrm{N}_2} = \mathrm{e}^{(\lambda_2-\lambda_1)} = \frac{1}{\mathrm{e}}$

$\Rightarrow \quad \mathrm{t} = \frac{1}{\lambda_2-\lambda_1} = 2$

Q.4 Answer:

$1 : 4$

Solution:

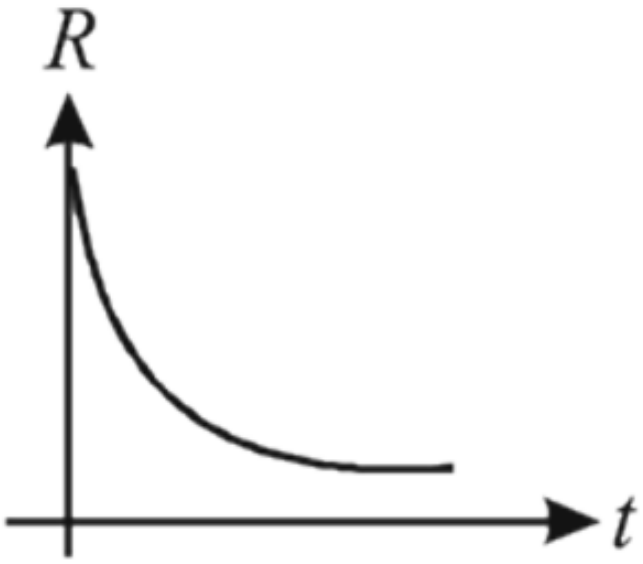
Ratio of number of half life taken is given as:

After 16 days $T_{A1/2} = \frac{16}{4} = 4; \quad T_{B1/2} = \frac{16}{8} = 2$

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$N = N_0 \left(\frac{1}{2}\right)^n \Rightarrow \frac{N_A}{N_B} = \frac{1}{2^4} : \frac{1}{2^2} = 2^2 : 2^4$
$= 4 : 16, = 1 : 4$

Q.5 Answer:



Solution:

$N = N_0 e^{-\lambda t} \Rightarrow N_y = N_0 \left(1 - e^{-\lambda t}\right)$
Rate of formation of $Y = \frac{dN}{dt} = +\lambda N_0 e^{-\lambda t}$
$X \longrightarrow Y$
At $t = 0, R = \lambda N_0 \Rightarrow t = 0 \quad N_0$
$t = \infty, R = 0 \quad \Rightarrow t = t \quad N$
$N_y = N_0 \left(1 - e^{-\lambda t}\right)$

Q.6 Answer:

$\left|\ell \mathrm{n}\left(\frac{\mathrm{A}_2}{2\,\mathrm{A}_1}\right)\right|$

Solution:

$\mathrm{A}_1 = (\lambda \mathrm{N}_0)\, \mathrm{e}^{-\lambda t_1}$
$\mathrm{A}_2 = (\lambda 2\, \mathrm{N}_0)\, \mathrm{e}^{-\lambda \mathrm{t}_2}$
$\mathrm{t}_1 - \mathrm{t}_2 = \frac{1}{\lambda} \ln\left(\frac{\mathrm{A}_2}{2\,\mathrm{A}_1}\right) = \ln\left(\frac{\mathrm{A}_2}{2\,\mathrm{A}_1}\right)$

Q.7 Answer:

11,400 years

Solution:

$\frac{C_{14}}{C_{12}} = \frac{1}{4} = \left(\frac{1}{2}\right)^{t/5700}$
$\Rightarrow \frac{t}{5700} = 2 \Rightarrow t = 11400$ **years**

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Q.8 Answer:

24 years

Solution:

Effective halflife is calculated as $\frac{1}{\mathrm{T}}=\frac{1}{\mathrm{T}_1}+\frac{1}{\mathrm{T}_2}$
$\frac{1}{\mathrm{T}}=\frac{1}{16}+\frac{1}{48} \Rightarrow \mathrm{T}=12$ **years**
Time in which $\frac{3}{4}$ **will decay is 2 halflives** $=24$ **years**

Q.9 Answer:

$6.95 \mu \mathrm{sec}$

Solution:

Let $t=$ **time for the potential of metal sphere to rise by one volt. Now** $\beta-$ **particles emitted in this time**
$=\left(6.25 \times 10^{11}\right) \times t$
Number of β **-particles escaped in this time** $=(80 / 100) \times\left(6.25 \times 10^{10}\right) t=5 \times 10^{10} t$
$\therefore$ **Charge acquired by the sphere in** $t$sec. $Q=\left(5 \times 10^{10} t\right) \times\left(1.6 \times 10^{-19}\right)=8 \times 10^{-19} t \cdots$ (i)
($\because$ **emission of** β **-particle lends to a charge** e **on metal sphere) The capacitance** C **of a metal sphere is given by** $C=4 \pi \varepsilon_0 \times r$
$=\left(\frac{1}{9 \times 10^9}\right) \times\left(\frac{10^{-3}}{2}\right)=\frac{10^{-12}}{18}$ **farad ...(ii)**
we know that $Q=C \times V\{$ **Here** $V=1$ **volt** $\}$
$\therefore \quad\left(8 \times 10^{-9}\right) t=\left(\frac{10^{-12}}{18}\right) \times 1$
Solving it for t**, we get** $t=6.95 \mu$sec.

Q.10 Answer:

172 and 69

Solution:

$${}_2 \mathrm{A}^{180} \xrightarrow{\alpha} 70 \mathrm{A}_1^{176} \xrightarrow{\beta} 71 \mathrm{A}_2^{176}$$
$$\xrightarrow{\alpha} 69 \mathrm{A}_3{}^{172} \xrightarrow{\gamma} 69 \mathrm{A}_4^{172}$$

Q.11 Answer:

α, β, γ

Solution:

α**-particles are essentially helium ions, and these are Massive and are Poorly penetrating.**
β**-radiation are fast-moving electrons, and these are slightly more penetrating with respect to** α **-particles**

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γ-radiation are faster moving particles, on the photonic scale. And these have the most penetrating power. They require lead shielding of several mm for safe containment.

Q.12 Answer:

5 years

Solution:

$X \xrightarrow{\lambda_1} Y \xrightarrow{\lambda_2} Z$

At radioactive equilibrium, $(\lambda_X) \times (N_X) = (\lambda_Y) \times (N_Y)$

$\frac{\lambda_X}{\lambda_Y} = \frac{N_Y}{N_X} \frac{\left(t_{1/2}\right)_Y}{\left(t_{1/2}\right)_X} = \frac{N_Y}{N_X}$ **or** $\frac{\left(t_{1/2}\right)_Y}{1.6 \times 10^{10}}$

$= \frac{1}{3.2 \times 10^9} \left(t_{1/2}\right)_Y = 5$ **years**

Q.13 Answer:

between 4 and 6

Solution:

Let $\mathrm{N_0}$ **be the number of atoms of** X **at time** $\mathrm{t} = 0$

Then at $\mathrm{t} = 4\mathrm{hrs}$ **(two half lives)** $\mathrm{N_x} = \frac{\mathrm{N_0}}{4}$ **and** $\mathrm{N_y} = \frac{3\,\mathrm{N_0}}{4}$

$\therefore \mathrm{N_x}/\mathrm{N_y} = 1/3$

and at $\mathrm{t} = 6\mathrm{hrs}$ **(three half lives)**

$\mathrm{N_x} = \frac{\mathrm{N_0}}{8}$ **and** $\mathrm{N_y} = \frac{7\,\mathrm{N_0}}{8}$ **or** $\frac{\mathrm{N_x}}{\mathrm{N_y}} = \frac{1}{7}$

The given ratio $\frac{1}{4}$ **lies between** $\frac{1}{3}$ **and** $\frac{1}{7}$ **Therefore, t lies between 4 hrs and** $6\mathrm{hrs}$.

Q.14 Answer:

5.78 per sec

Solution:

at time $\mathrm{t} = 0; \mathrm{A_0} = \frac{\mathrm{dN_0}}{\mathrm{dt}}$

$\frac{\mathrm{A}}{\mathrm{A_0}} = \frac{\mathrm{dN/dt}}{\mathrm{dN_0/dt}} = \frac{3260}{6520}$

or $\lambda = \frac{2.303}{\mathrm{t}} \log \frac{\mathrm{A_0}}{\mathrm{A}} = \frac{2.303}{2 \times 60} \log 2$

$= 5.78$ **per sec**

Q.15 Answer:

after any time

Solution:

Because radioactivity is a spontaneous phenomenon.

Q.16 Answer:

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20000dps

Solution:

Activity of sample becomes 2500 from 5000 in 75 days therefore its half life is 75 days, so $\mathrm{R} = \frac{\mathrm{R}_0}{2^{\frac{150}{75}}} = 5000 \Rightarrow \mathrm{R}_0 = 5000 \times 4 = 20,000$

Q.17 Answer:

$3.125\mu g$

Solution:

Using
$\frac{\mathrm{M}}{\mathrm{M}_0} = \left(\frac{1}{2}\right)^{\mathrm{t/T}} = \left(\frac{1}{2}\right)^{120/24} = \left(\frac{1}{2}\right)^{5} = \frac{1}{32}$
So $\mathrm{M} = \frac{\mathrm{M}_0}{32} = \frac{0.1\mathrm{mg}}{32} = 3.125\mu g$

Q.18 Answer:

The energy spectrum of γ - decay is continuous.

Solution:

Energy spectrum of α & γ decay **are line spectrum and that of** β - particle **is continuous.f**

Q.19 Answer:

It is not possible

Solution:

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Q.20 Answer:

$\frac{t_1 t_2}{t_1 + t_2}$

Solution:

$\frac{t_1 t_2}{t_1 + t_2}$

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